Here is a clear, step-by-step instruction template you can copy, paste, and send to your friend along with your `firmware_factory.bin` file.

🚀 How to Flash the Firmware on your M5Stack Tab5

Hey! Here is the compiled firmware for the Tab5 board. Because this is a merged factory binary, it contains the entire operating environment (bootloader, partition maps, and the app itself). You only need to flash this single file.

The easiest way to install it is using your web browser—no coding tools or IDEs required.

📋 Prerequisites

1. **A compatible browser: You must use Google Chrome, Microsoft Edge, or Opera (Safari and Firefox do not support Web Serial).**
2. **A USB-C Cable: Connect the Tab5 directly to your computer.**

🛠️ Flashing Steps

1. **Open the Web Flasher:**
   Go to the official online flasher tool: espressif.github.io/esptool-js/
2. **Connect the Board:**

- **Click the blue Connect button on the webpage.**
- **A browser popup will appear. Select the serial port corresponding to your ESP32-P4 / Tab5 board and click Connect.**
  (Note: If the chip type isn't automatically detected as ESP32-P4, change the "Chip" dropdown manually to ESP32-P4).

3. **Load the Binary:**

- **In the first file slot row, look for the File column and click Choose File.**
- **Select the `firmware_factory.bin` file I sent you.**

4. **Set the Target Address:**

- **In that same row, look for the Offset box.**
- **Make sure it is set precisely to `0x0` (zero).**

5. **Wipe and Flash:**

- ***(Optional but recommended)* Click Erase Flash first to clear out any old configurations on your board.**
- **Click Program to begin uploading.**
  Keep an eye on the terminal logs at the bottom of the page. Once it hits 100% and says successfully written, tap the physical Reset button on your Tab5 to boot into the new app!

⚠️ Troubleshooting: What if it won't connect or flash?

Because the Tab5 uses a native built-in USB interface, sometimes computers fail to auto-reset the chip into firmware download mode. If the webpage displays a connection timeout error, manually force the chip into Bootloader Mode:

1. **Press and hold the physical Boot button on the Tab5.**
2. **Press and release the Reset button.**
3. **Release the Boot button.**
4. **Go back to Step 2 on the website and click Connect again.**

Would you like to include any specific setup instructions for the app itself—such as Wi-Fi credential configuration or micro-SD card file requirements—before sending this template over?